ECE 757, Homework 1

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1 Profiling and analysis

To estimate the potential gains of parallelizing `kymppitonni.c`, we used the `getCC()` function to measure the execution time of each game as well as the total execution time of a run with the following parameters:

- Games per cutoff: 50
- Minimum cutoff: 10
- Maximum cutoff: 300

This yielded the given measurements:

- $t_{total} = 4400319$ (cycles)
- $t_{parallelizable} = 4360000$ (cycles)

Using Amdahl's law, we can then estimate theoretical speedups for 1, 2, 4, 8, and 16 processors. The results are detailed in the graph below.

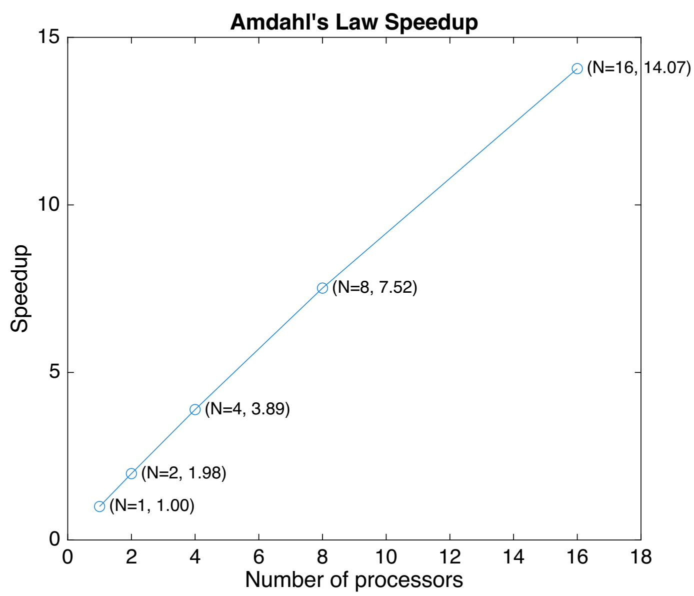

Figure 1: Theoretical Amdahl's Law Speedup

2 Running phtreads implementation on gem5 in SE mode